Prof. Shilpa Kastha

Theoretical Physics Division,
Saha Institute of Nuclear Physics
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Saltlake, Kolkata, India
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Date of birth: 9th August, 1990

RESEARCH INTERESTS

- Tests of General relativity with gravitational wave observations
- Post-Newtonian Theory
- Gravitational waveform modeling
- Blackhole spectroscopy
- Astrophysical implications of gravitational wave observations

EDUCATION

Ph.D (Theoretical Physics), The Institute of Mathematical Sciences, Chennai, India (2014–2019)

Thesis: Gravitational waves from compact binary coalescences: Tests of General Relativity and Astrophysics.
Advisors: Prof. K G Arun, Prof. Manjari Bagchi, Prof. Ghanshyam Date

MSc.(Physics, Specialization in Astrophysics & Cosmology), Visva Bharati, Shantiniketan, India (2012–2014)

Master's Thesis (@ IUCAA, India): Stochastic Gravitational Wave Background from a Set of Gravitationally Bound Massive Bodies,
Advisors: Prof. Sanjit Mitra, Prof. Biplab Raychaudhuri

B.Sc.(Honours) (Physics), Visva Bharati, Shantiniketan, India (2009–2012)

EMPLOYMENT

- Associate Professor E, Theoretical Physics Division at Saha Institute of Nuclear Physics(July, 2024–present day)
- Postdoc, Strong Gravity group at Niels Bohr International Academy, University of Copenhagen (Nov., 2022–June, 2024)
- Postdoc/Junior Scientist, MPI for Gravitational Physics (AEI) Hannover (Oct., 2019–Nov., 2022)

STUDENT & POSTDOC MENTORING

Postdocs: A. K. Mishra (2024-2026 at SINP)
Grad students: Mr. Saheb Das (currently at SINP)
Undergrad/masters students: Mr. Sai Karan M (2026-2027), Mr. Harshal Nitin Chaware (2025-2026), Mr. Zaryab Ahmed (2022-2023, at Niels Bohr Institute)

- Pycbc
- LalSuite
- GW ringdown packages: QNM, Pykerr, Ringdown
- MOTSFinder

PUBLICATIONS

1. **Cusp Formation in Merging Black Hole Horizons**
   Shilpa Kastha, Stamatis Vretinaris, Daniel Pook-Kolb, Badri Krishnan
   arXiv:2605.10874v2 [gr-qc], Submitted to Phys. Rev. D

2. **Physical significance of the black hole quasinormal mode spectra instability**
   Vitor Cardoso, Shilpa Kastha, Rodrigo Panosso Macedo
   Phys. Rev. D 110, 024016, arXiv:2404.01374v1[gr-qc]

3. **Constraining parametric deviation from Kerr using black hole ringdown of GW150914 and GW190521**
   Zaryab Ahmed, **Shilpa Kastha**, Alex B. Nielsen
   arXiv:2401.06049v1[gr-qc]

4. **Multiparameter multipolar test of general relativity with gravitational waves**
   Parthapratim Mahapatra, **Shilpa Kastha**, Anuradha Gupta, B. S. Sathyaprakash, and K. G. Arun
   Phys.Rev.D 109, 064036, arXiv:2312.06444v1[gr-qc]

5. **Parametrized multipolar gravitational waveform for testing general relativity: Amplitude corrections upto 2PN order**
   Parthapratim Mahapatra, **Shilpa Kastha**
   Phys.Rev.D. 109, 084069, arXiv:2311.04672[gr-qc]

6. **Gating-and-inpainting perspective on GW150914 ringdown overtone: Understanding the data analysis systematics**
   Yi-Fan Wang, Collin D. Capano, Jahed Abedi, **Shilpa Kastha**, Badri Krishnan, Alex B. Nielsen, Alexander H. Nitz, Julian Westerweck
   Phys. Rev. D 112, 083023, arXiv:2310.19645 [gr-qc]

7. **Spectroscopy for asymmetric binary black hole mergers**
   Jahed Abedi, Collin D. Capano,**Shilpa Kastha**, Yi-Fan Wang, Julian Westerweck, Alex B. Nielsen, Badri Krishnan
   Phys. Rev. D 108, 104009, arXiv:2309.03121 [gr-qc],

8. **Estimating false alarm rates of sub-dominant quasi-normal modes in GW190521**
   Collin D. Capano, Jahed Abedi, **Shilpa Kastha**, Alexander H. Nitz, Julian Westerweck,Yi-Fan Wang, Miriam Cabero, Alex B. Nielsen, Badri Krishnan
   Class. Quantum Grav. 41 245009, arXiv:2209.00640 [gr-qc]

9. **4-OGC: Catalog of gravitational waves from compact-binary mergers**
   Alexander H. Nitz, Sumit Kumar, Yi-Fan Wang, **Shilpa Kastha**, Shichao Wu, Marlin Schäfer, Rahul Dhurkunde, Collin D. Capano
   The Astrophysical Journal, Volume 946, Number 2, arXiv:2112.06878[astro-ph.HE]

10. **Model systematics in time domain tests of binary black hole evolution**
Shilpa Kastha, Collin D. Capano, Julian Westerweck, Miriam Cabero, Badri Krishnan, Alex B. Nielsen
Phys.Rev.D 105.064042, arXiv:2111.13664[gr-qc]

11. **Linear momentum flux from inspiralling compact binaries in quasi-elliptical orbits at 2.5 Post-Newtonian order**
Shilpa Kastha
Phys.Rev.D 105.064039 , arXiv:2110.12807[gr-qc]

12. **3-OGC: Catalog of gravitational waves from compact-binary mergers**
Alexander H. Nitz, Collin D. Capano, Sumit Kumar, Yi-Fan Wang, **Shilpa Kastha**, Marlin Schäfer, Rahul Dhurkunde, Miriam Cabero
The Astrophysical Journal, Volume 922, Number 1, arXiv:2105.09151[gr-qc]

13. **Multimode Quasinormal Spectrum from a Perturbed Black Hole** Collin D. Capano, Miriam Cabero, Julian Westerweck, Jahed Abedi, **Shilpa Kastha**, Alexander H. Nitz, Alex B. Nielsen, Badri Krishnan
Phys. Rev. Lett. 131, 221402, arXiv:2105.05238[gr-qc]

14. **Tests of general relativity using multiband observations of intermediate mass binary black hole mergers**
Sayantani Datta, Anuradha Gupta, **Shilpa Kastha**, K. G. Arun, B. S. Sathyaprakash
PhysRevD.103.024036, arXiv:2006.12137[gr-qc]

15. **Multiparameter tests of general relativity using multiband gravitational-wave observations**
Anuradha Gupta, Sayantani Datta, **Shilpa Kastha**, Ssohrab Borhanian, K. G. Arun, B. S. Sathyaprakash
PhysRevLett.125.201101, arXiv:2005.09607[gr-qc]

16. **Testing the multipole structure and conservative dynamics of compact binaries using gravitational wave observations: The spinning case**
Shilpa Kastha, Anuradha Gupta, K G Arun, B S Sathyaprakash, Chris Van Den Broeck
PhysRevD.100.044007, arXiv:1905.07277 [gr-qc]

17. **Testing the multipole structure of compact binaries using gravitational wave observations**
Shilpa Kastha, Anuradha Gupta, K G Arun, B S Sathyaprakash, Chris Van Den Broeck
PhysRevD.98.124033, arXiv:1809.10465 [gr-qc]

18. **Imprints of the redshift evolution of double neutron star merger rate on the signal-to-noise ratio distribution**
Shilpa Kastha, M. Saleem, K G Arun
MNRAS, Volume 496, Issue 1, July 2020, Pages 523–531, arXiv:1801.05942 [gr-qc]

19. **Stochastic Gravitational Wave Background from Exoplanets**
Anirban Ain, **Shilpa Kastha**, and Sanjit Mitra
PhysRevD.91.124023, arXiv:1504.01715 [gr-qc]

20. **Prospects for Fundamental Physics with LISA**
Enrica Barausse, Emanuele Berti, and others
Gen. Rel. Grav. volume 52, number 8, pages 81, year 2020, arXiv:2001.09793 [gr-qc]

INVITED TALKS

- Fine tuning the theoretical modeling of compact binaries
  33rd Indian Association for General Relativity and Gravitation, Bits Pilani, India (2025)
- Slipping into Inspiral: Gravitational Wave Emission from Compact binary
  Wednesday Weekly Colloquium, School of Astrophysics, Presidency University, , Kolkata, India (2025)
- Tests of general relativity with the gravitational wave observations
  Second Chennai Symposium on Gravitation and Cosmology, Indian Institute of Technology, Chennai, India (2022)
- Detection of a sub-dominant quasi-normal mode in GW190521
  Indian Institute of Sciences, Bengaluru, India (2022)
- 3-OGC: Catalog of gravitational waves from compact-binary mergers
  University of Stavanger, NO-4036 Stavanger, Norway (2021)
- Parameterized tests of the multipolar structure of compact binary inspiral
  Chennai Mathematical Institute, India (2020).
- Testing post-Newtonian multipolar structure of compact binaries using gravitational wave observation
  MPI (AEI), Hannover (2020)
- Parametrized tests of the multipole structure of compact binaries using gravitational wave observations : Spinning Case
  GR22 and Amaldi13, Valencia, Spain (2019)
- Testing the multipolar structure of compact binary spacetimes
  XXXVI Meeting of Astronomical Society of India, 2018, Osmania University, India

WORKSHOPS ATTENDED AND POSTERS PRESENTED

- **Niels Bohr International Academy Day 2023**
  Poster: *Tests of General Relativity with gravitational wave observations*
  Niels Bohr Institute, University of Copenhagen at the Odd Fellow Palace, February 9, 2023
- **The Gravitational Wave Physics and Astronomy Workshop (GWPAW 2021)**
  Poster: *Time domain test of binary black hole evolution*
  Max Planck Institute for Gravitational Physics (AEI), Hannover, Germany, December 14-17, 2021
- **Physics and Astrophysics at the eXtreme**
  IUCAA, Pune, India, August 7-10, 2018
- **XXXVI Meeting of Astronomical Society of India (ASI 2018)**
  Osmania University, Hyderabad, India, February 5-9, 2018
- **29th meeting of IAGRG**
  Poster: *Gravitational recoil of eccentric compact binaries*
  IIT Guwahati, Asam, India, May 18-20, 2017
- **The Future of Gravitational Wave Astronomy (FGWA 2016)**
  ICTS, Bangalore, India, May 18-20 2016
- **XXXIV Meeting of Astronomical Society of India (ASI 2016)**
  Poster: *Tracking the merger rate density evolution of double Neutron star binaries*
  University of Kashmir, Srinagar, India, May 10-13, 2016
- **8th International Conference on Gravitation and Cosmology (ICGC 2015)**
  Poster: *Probing the inhomogeneity in the spatial distribution of Double Neutron Star mergers Through Gravitational wave observations*
  IISER Mohali, India, December 14-18, 2015

TEACHING

- **Course Instructor** for **"General Relativity 1"**
  Course offered in *Saha Institute of Nuclear Physics*, SINP 2024
- **Tutor** for **"Gravitational waves from neutron star"**
  Course offered in *Summer School on Gravitational-wave Astronomy*, ICTS, 2017 by Nils Andersson
- **Tutor** for **"Dynamic formation of compact binaries in dense stellar clusters"**
  Course offered in *Summer School on Gravitational-wave Astronomy*, ICTS, 2018 by Sourav Chatterjee

CONFERENCE WORKSHOP ORGANIZATION

- **LISA in Copenhagen, Fundamental Physics with LISA**,
  Niels Bohr Institute, Copenhagen, 9–11 August 2023
- **Black hole inside out**
  Niels Bohr Institute, Copenhagen, 2024

SCHOOL ATTENDED

- **Summer school on gravitational wave astronomy**
  ICTS, Bangalore, India, August 13-24, 2018
- **Summer school on gravitational wave astronomy**
  ICTS, Bangalore, India, July 17-28, 2017
- **Summer school on gravitational wave astronomy**
  ICTS, Bangalore, India, July 25-August 5, 2016

PROFESSIONAL MEMBERSHIPS

- **Member of LISA consortium (July 2018-Present)**
- **Member of Gravitational Wave International Commitee 3G (March 2018-Present)**
- **Member of Indian Initiative in Gravitational-wave Observations (IndIGO) (Jun 2015-Present)**

RESEARCH IN THE NEWS

- "Machine learning splines bridge the final crash of colliding black holes," The Primary, 12 October 2026. Feature article on our research on machine-learning-assisted inspiral–merger–ringdown gravitational-wave modelling, including an interview with the author. Read article.